$$C_{lq}^{(3)}\,i1\_i2\_i3\_i4\_ \to$$

$$\frac{1}{16\pi^2}\Big(\frac{1}{9}\,g_2^4\,\mathrm{LF}_{3,0}[m_2]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{6}\,g_2^4\,\mathrm{LF}_{4,-1}[m_2]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{8}{45}\,g_2^4\,\mathrm{LF}_{5,-2}[m_2]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\sum_p g_2^4\,\mathrm{LF}_{3,0}\big[m_{\tilde{l}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{5}{48}\sum_p g_2^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde{l}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{2}{45}\sum_p g_2^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{l}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{6}\sum_p g_2^4\,\mathrm{LF}_{3,0}\big[m_{\tilde{q}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{16}\sum_p g_2^4\,\mathrm{LF}_{4,-1}\big[m_{\tilde{q}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{2}{15}\sum_p g_2^4\,\mathrm{LF}_{5,-2}\big[m_{\tilde{q}}^{p}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_2^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{12}\,g_2^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{4}{45}\,g_2^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{432}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{432}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_2,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,2,-1}\big[m_2,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{48}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_2,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{32}\,g_2^4\,\mathrm{LF}_{4,1,-2}\big[m_2,m_{\tilde{l}}^{i1}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_2,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,2,-1}\big[m_2,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{48}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_2,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{32}\,g_2^4\,\mathrm{LF}_{4,1,-2}\big[m_2,m_{\tilde{l}}^{i2}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_2,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,2,-1}\big[m_2,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{48}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_2,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{32}\,g_2^4\,\mathrm{LF}_{4,1,-2}\big[m_2,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_2,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{2,2,-1}\big[m_2,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{5}{48}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_2,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{32}\,g_2^4\,\mathrm{LF}_{4,1,-2}\big[m_2,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_2^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_2^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{l}}^{i1},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{l}}^{i1},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{l}}^{i1},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{l}}^{i1},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{l}}^{i2},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{96}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{l}}^{i2},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{l}}^{i2},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{l}}^{i2},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{432}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i3},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i3},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i3},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i3},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i3},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i3},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{432}\,g_1^2\,g_2^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i4},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{864}\,g_1^2\,g_2^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i4},m_1\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{48}\,g_2^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i4},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{96}\,g_2^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i4},m_2\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{9}\,g_2^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde{q}}^{i4},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}+\frac{1}{18}\,g_2^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde{q}}^{i4},m_3\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{8}\,g_2^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{3,1,-1}\big[\tilde{\mu},m_{\tilde{d}}^{p}\big]\,\delta_{i1i2}+\frac{1}{24}\,g_2^2\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\mathrm{LF}_{4,1,-2}\big[\tilde{\mu},m_{\tilde{d}}^{p}\big]\,\delta_{i1i2}-$$

$$\frac{1}{8}\,g_2^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{3,1,-1}\big[\tilde{\mu},m_{\tilde{e}}^{p}\big]\,\delta_{i3i4}+\frac{1}{24}\,g_2^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\mathrm{LF}_{4,1,-2}\big[\tilde{\mu},m_{\tilde{e}}^{p}\big]\,\delta_{i3i4}-$$

$$\frac{1}{8}\,g_2^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{3,1,-1}\big[\tilde{\mu},m_{\tilde{u}}^{p}\big]\,\delta_{i1i2}+\frac{1}{24}\,g_2^2\,\overline{y_u}^{i4p}\,y_u^{i3p}\,\mathrm{LF}_{4,1,-2}\big[\tilde{\mu},m_{\tilde{u}}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{1}{16}\,g_2^4\,\mathrm{LF}_{2,1,1,-1}\big[m_2,m_{\tilde{l}}^{i1},m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{8}\,g_2^4\,m_2^2\,\mathrm{LF}_{2,1,1,0}\big[m_2,m_{\tilde{l}}^{i1},m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{16}\,g_2^4\,\mathrm{LF}_{2,1,1,-1}\big[m_2,m_{\tilde{l}}^{i2},m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-\frac{1}{8}\,g_2^4\,m_2^2\,\mathrm{LF}_{2,1,1,0}\big[m_2,m_{\tilde{l}}^{i2},m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}+$$

$$\frac{1}{8}\,\overline{y_d}^{i4p}\,y_d^{i3p}\,\overline{y_e}^{i2r}\,y_e^{i1r}\,\mathrm{LF}_{2,1,1,-1}\big[\tilde{\mu},m_{\tilde{d}}^{p},m_{\tilde{e}}^{r}\big]-\frac{1}{4}\,\tilde{\mu}^2\,\overline{y_e}^{i2p}\,y_e^{i1p}\,\overline{y_u}^{i4r}\,y_u^{i3r}$$

$$\mathrm{LF}_{2,1,1,0}\big[\tilde{\mu},m_{\tilde{e}}^{p},m_{\tilde{u}}^{r}\big]-\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{1,1,1,1,-1}\big[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{24}\,m_1\,m_2\,g_1^2\,g_2^2\,\mathrm{LF}_{1,1,1,1,0}\big[m_1,m_2,m_{\tilde{l}}^{i1},m_{\tilde{q}}^{i3}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{48}\,g_1^2\,g_2^2\,\mathrm{LF}_{1,1,1,1,-1}\big[m_1,m_2,m_{\tilde{l}}^{i2},m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}-$$

$$\frac{1}{24}\,m_1\,m_2\,g_1^2\,g_2^2\,\mathrm{LF}_{1,1,1,1,0}\big[m_1,m_2,m_{\tilde{l}}^{i2},m_{\tilde{q}}^{i4}\big]\,\delta_{i1i2}\,\delta_{i3i4}\Big)$$